

THEORY cheat-sheet — Computational Finance (WI4151)

One condensed block per lecture: **core idea, key results, the line of reasoning**. No derivations — those live in notes/Lecture<NN>-notes.pdf. COS (L6, L10) is the centre of gravity.

Notation: $x = \ln(S/K)$ or $\ln S$ (log-state); φ = characteristic function (cf); $[a, b]$ = COS truncation range; Σ' = first term halved; cumulants c_1, c_2, c_4 ; $N(\cdot)$ = standard normal CDF.

Cluster 1 — Foundations / lattice / PDE (L1-L3, L5)

Lecture 01 — Asset classes, no-arbitrage, the Wiener process

- **Core:** the pricing primitives — discounting, no-arbitrage replication, and the stochastic model of S .
- **Key results:** bank account $M(t) = e^{rt}$, discount $e^{-r(T-t)}$; covered interest-rate parity $(1+i_s) = (F/S)(1+i_c)$; payoffs $(S_T - K)^+$, $(K - S_T)^+$ (kink at K); one-step hedge $\Delta = \Delta V / \Delta S$, price $= E^Q[\text{payoff}]$ (physical p irrelevant); put-call parity $C + Ke^{-r(T-t)} = P + S$; bound $C \geq (S - Ke^{-r(T-t)})^+ = \text{American call} = \text{European}$; GBM $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$, $\ln(S_T/S_0) \sim N((\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T)$; Wiener: $E[dW] = 0$, $(dW)^2 = dt$, $[W]_t = t$.
- **Reasoning:** price = cost of the replicating hedge $\Rightarrow$ risk-neutral expectation; $(dW)^2 = dt$ forces Itô calculus; GBM is the baseline the whole course perturbs.

Lecture 02 — Itô, the BS PDE, risk-neutral valuation, Feynman-Kac

- **Core:** three equivalent views of the European price — hedging PDE, Q -expectation, Feynman-Kac bridge.
- **Key results:** Itô $dg = (g_t + \mu g_x + \frac{1}{2}\sigma^2 g_{xx})dt + \sigma g_x dW$; BS PDE $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0$; under Q drift $\mu \rightarrow r$, σ unchanged; discounted price $e^{-rt}V$ is a Q -martingale; closed form $C = S_0 N(d_1) - Ke^{-rT} N(d_2)$, $d_{\{1,2\}} = [\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T] / (\sigma\sqrt{T})$, $N(d_2) = Q(S_T > K)$.
- **Reasoning:** delta-hedge removes $dW \Rightarrow$ portfolio earns $r \Rightarrow$ PDE; μ cancels in hedging, so the real drift never enters the price.

Lecture 03 — Implied volatility

- **Core:** invert BS price for the one unknown σ ; read the market's view off option prices.
- **Key results:** four knowns (S, K, T, r) , unknown σ ; vega $= S\sqrt{T} \varphi(d_1) > 0 \Rightarrow$ price strictly increasing in $\sigma = \text{unique IV}$; admissible price interval bounds existence; **bisection** (linear, robust) vs **Newton** (quadratic, divide-by-vega blows up deep OTM) $\Rightarrow$ globalise by seeding at the price-curve inflection point; put and call share one IV (parity); smile/skew in K , term structure in T .
- **Reasoning:** monotonicity (vega > 0) guarantees a unique root; a non-flat IV surface is the empirical proof that constant- σ BS is wrong — motivates Heston/jumps.

Lecture 05 — Binomial trees & finite differences

- **Core:** two lattice/grid pricers and the fact that a binomial tree is an explicit FD scheme.
- **Key results:** CRR u, d, p from matching mean+variance of the log-return to BS; risk-neutral backward induction; convergence $O(1/M)$, non-monotone (strike-straddling). FD on the heat equation: FTCS (explicit, stable only if $v = k/h^2 \leq \frac{1}{2}$), BTCS (implicit, unconditionally stable), Crank-Nicolson ($O(k^2 + h^2)$); von Neumann amplification $\xi_{\text{FTCS}} = 1 - 4v \sin^2(\beta h/2)$; BS $\rightarrow$ heat via $x = \ln S$, $V = e^{-r\tau} W$; binomial = FTCS with the $h^2 = \sigma^2 k$ coupling collapsing the centre node.
- **Reasoning:** explicit schemes are cheap but the CFL bound forces $k = O(h^2)$; work in $x = \ln S$ for constant coefficients and to tame the large- S diffusion weight (same domain-truncation lesson as the COS $[a, b]$).

Cluster 2 — Monte Carlo (L4)

Lecture 04 — MC integration, SDE simulation, variance reduction

- **Core:** unbiased sampling estimator with dimension-independent $O(N^{-1/2})$ error, plus how to make it faster.
 - **Key results:** $\text{Var}(\bar{X}_N) = \sigma^2/N$, $\text{SE} = s_N/\sqrt{N}$, CLT confidence interval; QMC reaches $\sim (\log N)^d/N$ (Koksma-Hlawka: error $\leq$ discrepancy $\times$ variation); Euler-Maruyama (weak 1, strong $1/2$), Milstein adds $\frac{1}{2}bb'[(\Delta W)^2 - h]$ (strong 1); variance reduction — antithetic ($\pm Z$), control variate (optimal $c^* = \text{Cov}/\text{Var}$, factor $(1-\rho^2)$), importance sampling (drift shift / likelihood ratio).
 - **Reasoning:** MC wins in high dimension because the rate is d -free; it is the slow benchmark that COS/FFT beat for smooth low-dimensional European problems.
-

Cluster 3 — Fourier / COS (priority) (L6, L10)

Lecture 06 — The Fourier family & the COS method

- **Core:** compute $e^{-rT}E[\text{payoff}]$ using only the cf, by expanding the density in a cosine series whose coefficients come straight from ϕ .
- **Key results:** cf $\phi_X(t) = E[e^{itX}]$ is the Fourier dual of the density; density coefficients $A_k = (2/(b-a))\text{Re}\{\phi(k\pi/(b-a))e^{-ik\pi a/(b-a)}\}$ (only approximation = range truncation); **European COS price** $P \approx e^{-rT} \sum_{k=0}^{N-1} \text{Re}\{\phi(k\pi/(b-a); x_0) e^{-ik\pi a/(b-a)}\} V_k$; payoff coefficients V_k closed-form via χ_k, ψ_k (call $(2K/(b-a))[\chi_k(\theta, b) - \psi_k(\theta, b)]$); **cumulant range** $[a, b] = c1 \pm L\sqrt{(c2 + \sqrt{c4})}$, $L \approx 8-12$; cost $O(N)$, **no FFT, no quadrature**; exponential convergence for smooth densities (saturates at the range floor).
- **Reasoning:** cosine basis $\Rightarrow$ real coefficients read off ϕ in one line, and V_k closes in elementary functions; the only error you control is $[a, b]$ (range) and N (series). **Short-maturity trap:** $c2 = \sigma^2 T \rightarrow 0$ collapses $[a, b]$; error then grows as $T \rightarrow 0$ and is *not* cured by more terms — fix by raising L / flooring the half-width / covering the whole grid (the Assignment-2 bug).

Lecture 10 — Barrier options

- **Core:** payoff $\times$ survival indicator; analytic under GBM, and a COS *backward recursion* for discrete monitoring.
 - **Key results:** 8 types (up/down $\times$ in/out $\times$ call/put); in-out parity $C_{\text{in}} + C_{\text{out}} = \text{vanilla}$; expectation form with $1_{\{\tau_B > T\}}$ vs localized BS PDE with **absorbing** boundaries $V(a) = V(b) = 0$. GBM analytics rest on the **reflection principle** (max density $p_{\Xi}(a) = \sqrt{(2/\pi T)}e^{-a^2/2T}$), joint density of (W_T, Ξ) , giving closed-form barrier prices; PDE routes = method of images $U = S^{2\alpha}V(B^2/S)$ and a sine-series heat solution. **Discrete COS:** Markov = the survival product factorises $\Rightarrow$ backward recursion; at each monitoring date $\hat{c} = e^{-r\Delta t} \sum F_k(x) V_k$, then knock-out splits V_k at the barrier h into a payoff piece (Ψ_k) and a continuation piece ($\hat{C}_k$); the coupling $M_{\{k, j\}}$ is **Hankel + Toeplitz** $\Rightarrow$ each step is a convolution done by **FFT**; total cost $O((M-1)N \log_2 N)$.
 - **Reasoning:** the barrier enters only through the boundary/indicator, not the dynamics. Key contrast: European COS uses no FFT, but the discrete barrier must propagate all $\{V_k\}$ every step — a structured matrix-vector product — so the FFT returns.
-

Cluster 4 — Models (L7, L8)

Lecture 07 — Heston stochastic volatility

- **Core:** add a mean-reverting variance with its own Brownian shock; price by plugging the affine cf into COS.

- **Key results:** $dS = rS dt + \sqrt{v} S dW_1$, $dv = \kappa(\bar{v} - v)dt + \eta\sqrt{v} dW_2$, $d(W_1, W_2) = \rho dt$; CIR variance $\sim$ noncentral χ^2 ; **Feller** $2\kappa\bar{v} \geq \eta^2$ keeps $v > 0$. Not affine in (S, v) but **is** affine in $(\ln S, v)$ $\Rightarrow$ cf $\phi = \exp(\alpha(\tau, u) + \beta(\tau, u)v + iu \ln S)$ from **Riccati ODEs** (closed form with $D = \sqrt{(\kappa - i\rho\eta)^2 + \eta^2(u^2 + iu)}$), G); the **little-Heston trap** (branch cut of complex $\log/\sqrt{\cdot}$) fixed by the right grouping. Simulation: Euler can make $v < 0$ (truncation/reflection/ $\log-v$ fixes); **Broadie-Kaya** exact via noncentral- $\chi^2 v$ + cf of integrated variance.
- **Reasoning:** Heston is the headline COS use case — one analytic cf prices the whole smile; $\rho < 0$ and small c_2 make the short-maturity range trap worse.

Lecture 08 — Jumps and affine jump-diffusion

- **Core:** add discontinuities; the cf still closes, so COS still works.
- **Key results:** Poisson $P(N=k) = e^{-\lambda t} (\lambda t)^k / k!$, $\text{mean} = \text{var} = \lambda$; compensated $N_t - \lambda t$ is a martingale; **compound-Poisson** cf $\exp(t\lambda(v(u) - 1))$ (Lévy-Khintchine exponent); Merton drift correction $\mu = r - \lambda E[J]$ (compensator); **Merton** (lognormal-jump) and **Kou** (double-exponential, $\eta_1 > 1$) cfs; non-local **PIDE**; **AJD** (Duffie-Pan-Singleton) cf $= \exp(\alpha + \beta \cdot x)$ from Riccati ODEs with the $(\theta(\beta) - 1)$ jump-transform term; BS recovered as a no-jump AJD.
- **Reasoning:** the $(\theta - 1)/\text{compensator}$ motif recurs (compensated Poisson, drift correction, AJD ODE); the only thing COS needs from any of these models is ϕ evaluated at $k\pi/(b-a)$ — heavy tails ($c_4 > 0$) widen $[a, b]$.

Cluster 5 — American options (L9)

Lecture 09 — Early exercise

- **Core:** optimal stopping / free boundary; price the Bermudan by backward recursion, extrapolate to American.
- **Key results:** continuation vs stopping region, smooth pasting at $S^*(t)$; **American call = European call** (non-dividend; dominating-strategy + parity bound), put breaks it ($Ke^{-r(T-t)} < K$); put as **LCP/obstacle** $(v - \Lambda) \geq 0$, $L_v \leq 0$, $(v - \Lambda)L_v = 0$. Methods on the shared $v = \max(\Lambda, c)$ recursion: binomial (one max line differs from European), FD + **PSOR** (project inside each sweep), **LSM** (regress discounted future CF on a basis over ITM paths; low-biased, foresight bias $\Rightarrow$ out-of-sample), **COS/Bermudan** (split V_k at the early-exercise point x_{n^*} found by Newton, C_k via FFT, $O(NK \log K)$); **Richardson** Bermudan $\rightarrow$ American.
- **Reasoning:** all four methods compute the same continuation value $E[v(t_{n+1}) | x]$ differently; the COS range caveat bites *per step* ($c_2 = \sigma^2 \Delta t$).

Cluster 6 — Exotics overview (L11)

Lecture 11 — Exotics map

- **Core:** survey of path-dependent / multi-asset payoffs and which numerical method fits each.
- **Key results:** **Asian** (avg) — arithmetic has no closed form (MC + geometric control variate; SV via Fouque-Han 3-D PDE, Vecer numeraire reduction 3D $\rightarrow$ 1D); **digital/binary** $= e^{-rT} N(d_2)$, but pin-risk / delta blow-up near K at expiry; **basket / worst-of** — correlation-driven, MC or moment-matching / 2-D COS; **cliquet** — forward-starting capped/floored returns (forward-skew sensitive); **barrier** — see L10.
- **Reasoning:** method follows structure — averaging/multi-asset/high-dimension $\Rightarrow$ MC; smooth low-dimensional with a known cf $\Rightarrow$ COS; the digital's discontinuous payoff is the recurring hedging-difficulty trap.

The through-line (say it in one breath)

Every price is $e^{-rT}E^Q[\text{payoff}]$. If the **density** is known (GBM) $\rightarrow$ closed form / PDE / trees. If only the **cf** is known (Heston, jumps, AJD) $\rightarrow$ **COS**: expand the density in a cosine series with coefficients $\text{Re}\{\phi \cdot e^{-ik\pi a/(b-a)}\}$, pair against closed-form payoff coefficients V_k , sum $O(N)$ terms; pick $[a,b]=c1 \pm L\sqrt{(c2+\sqrt{c4})}$ and watch the short-maturity collapse. Early exercise (L9) and discrete barriers (L10) reuse COS as a **backward recursion**; high-dimensional/averaging payoffs (L11) fall back to **Monte Carlo** (L4).